



Jérémie KALFON

Phone +33 652663443
Email jkobject@gmail.com
Website jkobject.com
LinkedIn www.linkedin.com/in/jkobject
Github www.github.com/jkobject
Address 10b rue montgallet, 75012 Paris, France

Profile

Computational Biologist, Researcher, Engineer, and team leader. Founder of PiPle, I am mission-driven and have a lifelong passion for Research and Science.

Highly independent, I am dedicated to improving human health using Artificial Intelligence. I am now finishing a Ph.D. and previously worked in research institutes and startup companies.

Skills

BigData: GCP, AWS, batch, SLURM, snakemake, terra, nextflow, HPC, ...

Biology: Systems biology, Cancer Biology, epigenomics, Neuroscience...

Datatype: single cell, RNAseq, ST, DNAseq, ChIP-seq, ATACseq, slam-seq, BS-seq, Perturb-seq, bi-photon imaging...

Code & frameworks: pytorch, python, javascript, java, R, matlab, firebase, SQL, Neo4j, nodeJS, react, ...

Datasets: CZI, HEST, CCLE, TCGA, GTEX, Gnomad, GWASatlas, ENCODE, Human Cell Atlas, ...

ML: Transformers, VAEs, GNNs, CNNs, NMF, OT, optimization...

Languages: English, French, Spanish

Leadership: team lead, project lead, project management, mentorship, leadership studies

Education

Pasteur Institute - ENS	PARIS, FR	2023-2026
Ph.D. Artificial Intelligence & Biology		
Harvard University	CAMBRIDGE, USA	2019-2021
Graduate Certificate in Biology		
University of Kent	CANTERBURY, UK	2017-2018
Master of Science. Machine Learning		
ECE Paris	PARIS, FR	2013-2018
Tec de Monterey	QUERETARO, MX	2014
French Diplôme d'ingénieur (M.sc.), Biomedical Engineering		

Experience

Advisor / Expertise 2023-2026

- advised multiple companies on scientific strategy - Nucleate Leader - member of various AI, Bio, Dev & VC communities

WhitelabGx - Team Lead in Computational Biology

- Cellular Atlas & Knowledge Graph for target discovery & cell surface marker discovery. AI for Cellular therapies. Hit Discovery.
- ML, BIG DATA, GENOMICS, STRUCURAL BIO 2022-2023

Broad Institute of MIT and Harvard - Senior Associate Computational Biologist (Project Lead)

- DepMap Omics, Predicting cancer dependencies from genetic sequences of cancer cell lines.
- ML, BIG DATA, GENOMICS 2019 - 2022
- AMLproject, deciphering AML from deep epigenetic profiling, with Dana Farber Cancer Institute & Boston Children's Hospital
- ML, BIG DATA, GENOMICS
- Celligner, Alignment of cancer models to patient tumors using conditional VAEs on multi omics data.
- ML, DL, GENOMICS 2021 -2022

PiPle - co-Founder - CEO

- Co-Founder of PiPle, A startup focused on bringing the next-generation communication platform. PiPle. INSEEC incubator
- PROJECT DEV, ENTREPRENEUR 2016 - 2023

Flatiron Institute, Simons Found. - Research Associate

- Member of the CalmAn project, Neural spike inference from calcium imaging. Flatiron Institute under the supervision of Dmitri Chklovskii
- RESEARCH, IMAGING, ML, VISION 2017

HighLife Med - R&D

- Stage in HighLife Med, A Parisian Biomedical Startup developing the world's first no-surgery mitral heart valve replacement.
- SIMULATION, DATA ANALYTICS 2016

Interests



Awards

- #1 AI in Bio Hackaton @ Entrepreneur First
- Best Poster Awards Pasteur Institute AI in Bio Seminar
- 2x Broad Spotlight Awards, For outstanding contribution. 2020, 2022.
- ECE innov' Awards, 2nd Place for the PiPle project, innovative Chat app introducing the concept of message-mapping. 2017
- Berger-Levrault Innovation Awards, 2nd Place for the PiPle project, innovative Chat app introducing the concept of message-mapping. 2017

Publications

Scholar: <https://scholar.google.com/citations?user=zyXiydsAAAAJ>

First Author

- *scPRINT-2: Towards the next-generation of cell foundation models and benchmarks.* J Kalfon, G Peyré, L Cantini. Submitted to Nature Methods
- *Towards foundation models that learn across biological scales.* J Kalfon, L Cantini, G Peyré, ICML 2025 FM4LS, Biorxiv.
- *scPRINT: pre-training on 50 million cells allows robust gene network predictions.* J. Kalfon, J. Samaran, G. Peyré, L. Cantini. Nature Communications. 2024.
- *Leukemia core transcriptional circuitry is a sparsely interconnected hierarchy stabilized by incoherent feed-forward loops.* J. Kalfon, T. Harada, Y Heshmati, BioRxiv, 2023.
- *A distinct core regulatory module enforces oncogene expression in KMT2A-rearranged leukemia,* J. Kalfon, T. Harada, Y Heshmati, Genes & Development, 2021. <https://doi.org/10.1101/2021.08.03.454902>

Highly Cited

- *CalmAn: an open source tool for scalable calcium imaging data analysis.* A Giovannucci, J. Friedrich, P. Gunn, J. Kalfon. ELife, 2019.
- *Repeat expansions confer WRN dependence in microsatellite-unstable cancers,* N. Wietmarschen, S. Sridharan, W. J. Nathan,..., Nature, 2020. <https://doi.org/10.1038/s41586-020-2769-8>
- *Synthetic Lethal Interaction between the ESCRT Paralog Enzymes VPS4A and VPS4B in Cancers Harboring Loss of Chromosome 18q or 16q,* J. E. Neggers, B. R. Paolella, A. Asfaw, M. V. Rothberg, Cancer Research, 2020. <https://doi.org/10.1016/j.celrep.2020.108493>

Last Author

- *Training foundation models on large collections of scRNA-seq data.* R. Sergei, F. Felix, W. Maciek, G. Ilan, R. Yanay, S. Sunny, S. Chaichontat, T. Fabian, J. Kalfon, A. Wolf

5 first author papers, 17 papers, 1200+ citations, 6 posters, 10 Conferences,